SPYRIDON MOUSOURIS, PHD

SOFTWARE DEVELOPER | GEODATA VISUALIZATION

PERSONAL INFORMATION

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+30 694 62 92 804 - LinkedIn

Athens, Greece

PROFESSIONAL EXPERIENCE

Front-End Developer at *Terra – Mapping the Globe* (12 / 2023 – 08 / 2026)

Developed large-scale web GIS apps for the private and the public sector

Full Stack Developer, Geodata Visualization at *Dipylon* (02 / 2022 – 10 / 2023)

Developed custom geovisualization and storytelling apps for archaeological and cultural data

Full Stack Developer at *Propulsion Analytics* (07 / 2019 – 02 / 2021)

Co-developed a custom web platform for visualizing vessel data and performance metrics

Academic Researcher at *University of Patras* (03 / 2017 – 06 / 2019)

Developed map-based solutions for cultural data

Front-End Developer at *Biblop* (05 / 2016 – 02 / 2017)

Developed custom, responsive front-end UIs

Academic Researcher at *University of Patras* (08 / 2012 – 01 / 2015)

Developed map-based solutions for cultural data

PROJECTS

Front-End Developer

Terra – Mapping the Globe
12 / 2023 – 08 / 2026

Single Digital Map (SDM)

https://sdigmap.tee.gov.gr/sdmportal/

- Developed the front-end to CRUD government issues and resolutions. Contains map-based functionality to find and project issues.
- Technologies: Typescript, React, React Router, React Query Builder, Axios, Zustand, ArcGIS Experience Builder, ArcGIS On Line

Cargo Tracking & Monitoring Platform

- Analyzed client requirements and researched existing solutions
- UX prototyping in Figma
- Developed demos for real-time geodata using React and ArcGIS Maps SDK
- Technologies: React, Typescript, Python, ArcGIS Maps SDK for JavaScript, ArcGIS Velocity, Figma

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Full Stack Developer
Data Geovisualization

Dipylon
02 / 2022 – 10 / 2023

Sikyon Excavation Map, first version

https://v1.sikyonexcavation.gr/

- Map-based web application for exploring excavation data through category-based filtering, spatial inclusion and interactive map navigation
- Designed a polygon-extrusion solution to visually distinguish selected areas and reduce overlapping
- Technologies: React, React Router, mapbox.js, turf.js, Axios, Node.js, express.js, Geoserver, PostgreSQL, PostGIS, QGIS, Figma

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Full Stack Developer

Propulsion Analytics
07 / 2019 – 02 / 2021

VesselQUAD

https://propulsionanalytics.com/VesselQUAD/

- Co-developed a custom web platform for visualizing vessel data and metrics.
- Developed APIs and front-end to retrieve and display data and algorithmic results.
- Implemented role-based access control to support different user categories and permission levels.
- Integrated WebSockets to stream large data volumes to feed data-driven charts
- Technologies: node.js, express.js, WebSockets, Vue.js, vuetify, Highcharts.js

Ikospot

Front-End Developer

Biblop

05 / 2016 – 02 / 2017

- Developed the front-end for a real-estate platform
- Translated design mockups into responsive web pages
- Implemented interactive UI elements and integrated front-end pages with PHP and MySQL back-end functionality.
- Technologies: HTML5, CSS3, JavaScript, jQuery, Responsive Web Design, PHP, MySQL, phpMyAdmin

TECHNICAL SKILLS

Programming Languages	SQL, JavaScript, Typescript, Python
Databases	PostgreSQL
Back-End	Node.js, Express.js
GIS & Geospatial	Geoserver, mapbox, PostGIS, QGIS
Front-End	React, React Router, Zustand
UX prototyping	Figma
AI-Assisted Development	Claude Code
Semantic Web & Knowledge Engineering	RDF, OWL, Protégé, Apache Jena Fuseki, SPARQL, SWIRL
Version Control Systems	Git
Containerization	Docker

EDUCATION

PhD in Cultural Informatics, (2019 – 2026), Department of Cultural Technology and Communication, University of the Aegean.
Developed a reference model to define and inference cultural states for points of interest

MSc 8.6 / 10, (2009 – 2011), Department of Cultural Technology and Communication, University of the Aegean.

BSc 8.3 / 10, (2005 – 2009), Cultural Heritage Management and New Technologies Department, University of Ioannina.

PUBLICATIONS AND CONFERENCE PAPERS

- Mousouris, S., Kavakli, E., Kotis, K. "*Definition and Evaluation of the Spatiotemporal States Reference Model (SStaRM) Ontology Design Pattern*", In: Journal on Computing and Cultural Heritage. 2026, Under Review
- Mousouris, S., Kavakli, E., Kotis, K. "*Defining Axioms to Address the Cultural Snapshot Phenomenon: Introducing the Spatiotemporal States Reference Model (SStaRM).*" In: Metadata and Semantic Research, MTSR 2025. Under Publication
- Mousouris, S., Kavakli, E. "*Towards a reference model for addressing the cultural snapshot phenomenon in cultural mapping libraries*". In: Int. J. Metadata, Semantics and Ontologies 2024, 17, 1. https://doi.org/10.1504/ijmso.2024.145509
- Mousouris, S., Kavakli, E. "*Towards Addressing the Cultural Snapshot Phenomenon in Cultural Mapping Libraries*." In: Metadata and Semantic Research. MTSR 2022. Communications in Computer and Information Science, 1789. Springer. https://doi.org/10.1007/978-3-031-39141-5_10
- Mousouris Spiridon, Styliaras Georgios, (2018) *"The Research Development of Cultural Mapping Libraries"*, IEEE - Information, Intelligence, Systems and Applications (IISA 2018)
- Mousouris Spiridon, Styliaras Georgios. *"On The Implementation of a Digital Map for Cultural Heritage"*. International Journal of Computational Intelligence Studies, Special Issue on: "Content Management and Dissemination in Multimedia Systems for Culture". InderScience Publishers. 2016 Vol. 5 No. 1, pages 30-49.
- Mousouris Spiridon, Styliaras Georgios, *"Implementing Digital Cultural Heritage Map"*, IEEE - Information, Intelligence, Systems and Applications (IISA 2014)
- Mousouris Spiridon, Styliaras Georgios, *"On the Creation of a Digital Cultural Heritage Map"*, IEEE - Information, Intelligence, Systems and Applications (IISA 2013)

LANGUAGES

English (advanced, CEFR C1), Greek (native)

ADDITIONAL INFORMATION

Valid Greek and Canadian passport. References available upon request.